



SEMANTIC LINK LABS: A LINK TO THE FUTURE

JASON ROMANS

INDY BI USER GROUP 2025

SEMANTIC LINK LABS

— A LINK TO THE FUTURE —



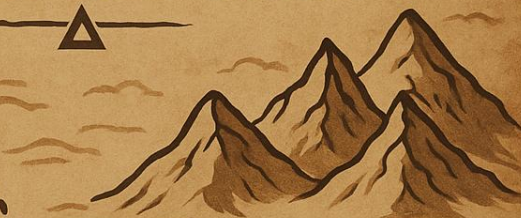
MICROSOFT
NOTEBOOK



REPORTS THAT
BREAK DUE TO
STRUCTURAL CHANGES



MODELS THAT
PERFORM POORLY
BECAUSE BEST
PRACTICES WERE
SKIPPED



USABILITY
PITFALLS THAT
MAKE REPORTS
“TECHNICALLY FINE”
BUT FUNCTIONALLY
BROKEN FOR
END USERS



GOVERNANCE
REPORT

JASON ROMANS

Cloud Data & Integration Developer

The DAX Shepherd 🤝



 @sql_jar

 jason-r-sql-jar

 <https://thedaxshepherd.com/>



 Nashville, TN, USA

 Began Career as a SQL Server DBA

 Transitioned to Microsoft BI Stack

 Data Engineering to Data Modeling

 Infrequent Blogger

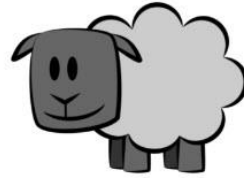
 Fan of Dimensional Models & Doctor

SHOULDERS OF GIANTS



SLIDES

www.thedaxshepherd.com



The DAX Shepherd

Musings on the Microsoft BI Stack



[Home](#) [About Me](#) [Simple Talk](#) [Presentations](#) [A Speaker's Journey](#)

Presentations

Sessionize

This is my [Sessionize Profile](#) that has the conferences I have spoken at along with future events. It has a couple of my most popular sessions.

Presentation Slides

This is my [GitHub Repository](#) with the presentation slides for each event.

Recorded Sessions

Simple Talks Podcast | Episode 4 – Coffee chat with Jason Romans

About Jason Romans



I love working with the Microsoft BI Stack. I am passionate about learning.

[A Speaker's Journey](#)

WHY GO ON THIS JOURNEY?

- ❖ LEVERAGE EXISTING SEMANTIC MODELS

- ❖ ADMINISTRATION

- ❖ DATA VALIDATION

 - ❖ Great Expectations (Python Library)

- ❖ BEST PRACTICES

- ❖ STAY AHEAD OF ISSUES

- ❖ “HAVE YOU TRIED SEMPY?”

OUR JOURNEY

1. INTRO
2. NOTEBOOKS
3. SEMANTIC LINK
4. SEMANTIC LINK LABS
5. CONCLUSION

OUR JOURNEY



1. INTRO

2. NOTEBOOKS

3. SEMANTIC LINK

4. SEMANTIC LINK LABS

5. CONCLUSION

MICROSOFT FABRIC ARCHITECTURE

- ❖ Need a Fabric Capacity
- ❖ Leverages Your Power BI Artifacts
- ❖ Data is stored in OneLake
- ❖ Choice of Compute engines

YOUR EXISTING SKILLS APPLY

- ❖ Build on your existing Jupyter Notebook experience
- ❖ Familiar Python environment
- ❖ Native Pandas support

OUR JOURNEY



1. INTRO

2. NOTEBOOKS

3. SEMANTIC LINK

4. SEMANTIC LINK LABS

5. CONCLUSION

TYPE OF NOTEBOOKS

LANGUAGE

- Python
- Scala
- Spark SQL
- R
- T-SQL

COMPUTE

- Spark
- Python (Single Node Virtual Machine)
- T-SQL Analytics



COMPUTE & LANGUAGE

 PySpark (Python)

Spark

✓ PySpark (Python)

Spark (Scala)

Spark SQL

SparkR (R)

Python

Python

T-SQL Analytics

T-SQL

SPARK (PYTHON, SCALA, SQL, R)

 PySpark (Python)

Spark

✓ PySpark (Python)

Spark (Scala)

Spark SQL

SparkR (R)

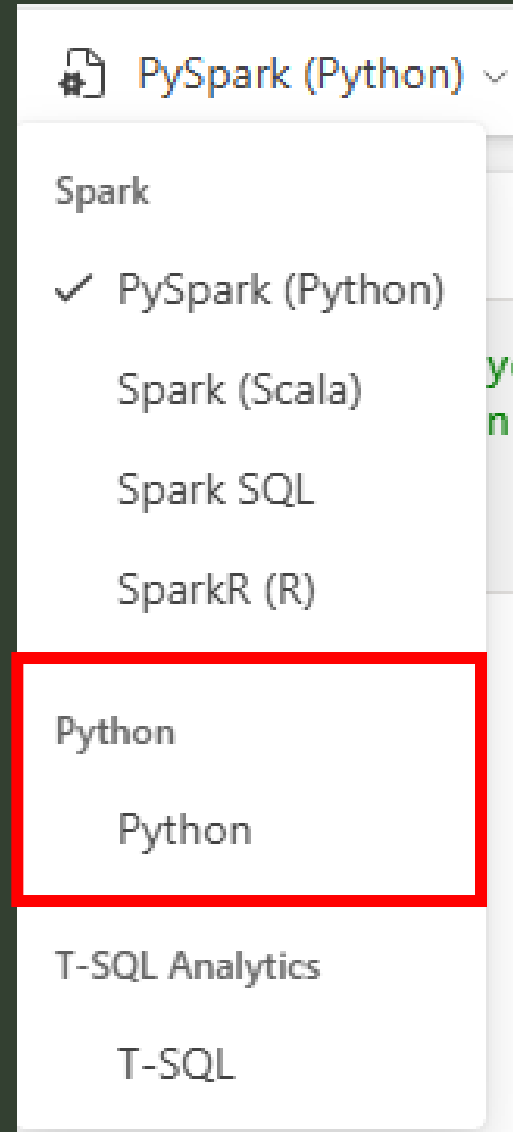
Python

Python

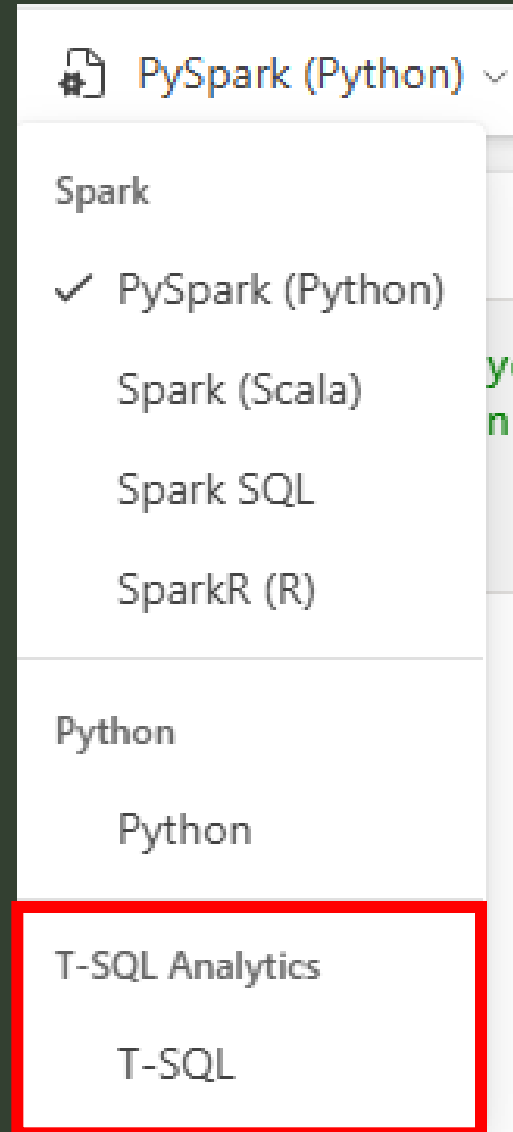
T-SQL Analytics

T-SQL

PYTHON (PYTHON)



T-SQL ANALYTICS (T-SQL)



CHOOSING PYSPARK (CLUSTER) OR PYTHON COMPUTE (SINGLE NODE)

Scenario	Recommended Notebook
Includes pre-installed DuckDB and Polars libraries	Python Notebooks
Small to medium data (fits in memory)	Python Notebooks (or PySpark on single-node Spark cluster)
Rapid exploration & prototyping	Python Notebooks (or PySpark on single-node Spark cluster)
Large datasets (10GB+) exceeding memory	PySpark Notebooks
Complex data workflows or ETL pipelines	PySpark Notebooks
High-concurrency or parallel execution	PySpark Notebooks
Needs Spark-native APIs (MLlib, SQL, Streaming)	PySpark Notebooks

<https://learn.microsoft.com/en-us/fabric/data-engineering/fabric-notebook-selection-guide>

TYPE OF COMPUTE FOR NOTEBOOKS

- Spark Based
 - Cluster
- Single Node Python
 - 2 vCores, 16G RAM (starter)
- T-SQL Analytics
 - Warehouse

PYTHON NOTEBOOK - CHANGE VM SIZE

NOTE: LONGER STARTUP TIMES

```
%%configure
```

```
{
```

```
    "vCores": 16
```

```
}
```

- Memory is proportional to vCores
 - 2 is 2vCores with 16G
 - 4 is 4vCores with 32G
 - 8 is 8vCores with 64G
 - 16 is 16vCores with 128G
 - 32 is 32vCores with 256G
 - 64 is 64vCores with 512G
 - (32 and 64) not available in trial

PYTHON NOTEBOOK

- Has libraries installed for dealing with “small-big” data
 - Less than 10 Gigabytes
 - Fits in memory
- Example Libraries installed
 - Polars
 - DuckDB

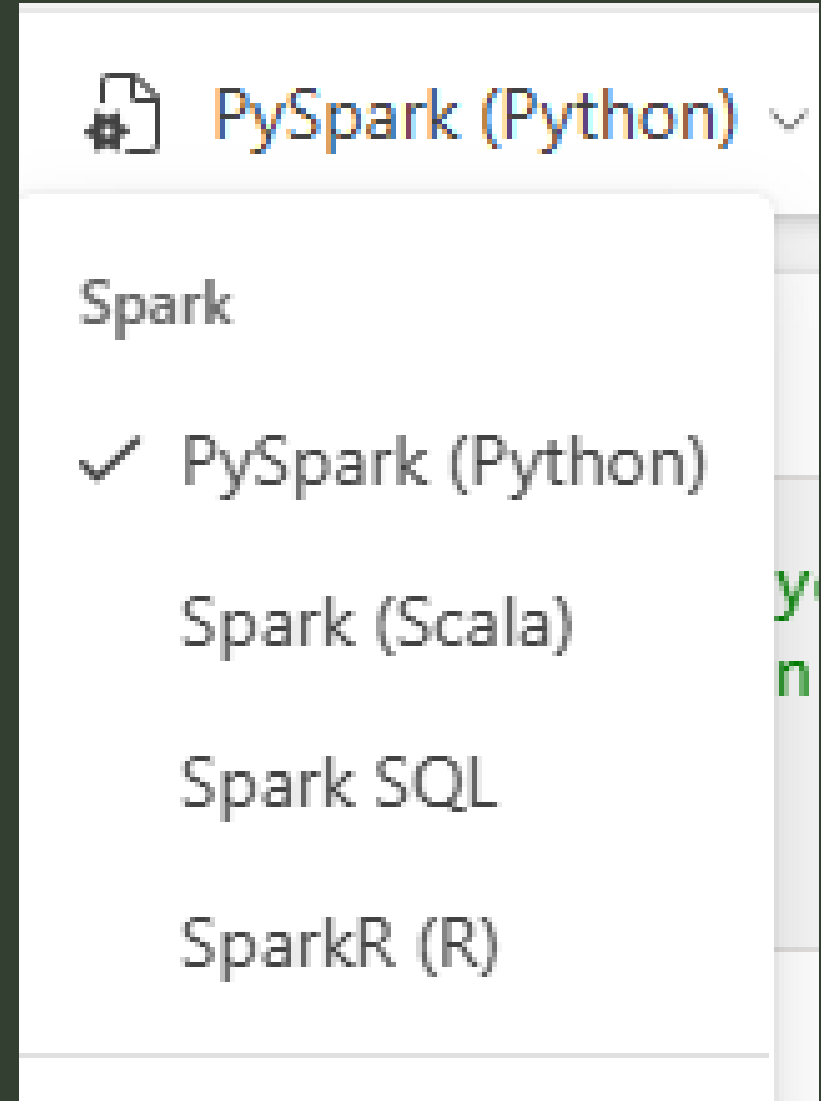
LANGUAGES FOR SPARK

Different choices of languages

Spark was built with Scala

- PySpark (Python)
- Spark (Scala)
- Spark SQL
- SparkR (R)

*Spark was built with Scala



MAGIC COMMANDS – SET LANGUAGE BY CELL

Magic command	Language	Description
%%pyspark	Python	Execute a Python query against Apache Spark Context.
%%spark	Scala	Execute a Scala query against Apache Spark Context.
%%sql	SparkSQL	Execute a SparkSQL query against Apache Spark Context.
%%html	Html	Execute a HTML query against Apache Spark Context.
%%sparkr	R	Execute a R query against Apache Spark Context.

Home

Edit

AI tools

Run

View

Comments

History

Develop

Share



Run all

Connect

PySpark (Python)

»

▼

Magic Commands

1

`%%spark`

Spark (Scala)

OUR JOURNEY



1. INTRO

2. NOTEBOOKS

3. SEMANTIC LINK

4. SEMANTIC LINK LABS

5. CONCLUSION

MISSING LINK

Semantic link is a feature that allows you to establish a connection between semantic models and Synapse Data Science in Microsoft Fabric.

-- Reference: <https://learn.microsoft.com/en-us/fabric/data-science/semantic-link-overview>

ALL THE THINGS

Semantic link is a feature that allows you to establish a connection between semantic models and Synapse Data Science in Microsoft Fabric.

semantic models

Models, Reports, Lakehouse, Workspaces and more

Synapse Data Science

Fabric Notebook – Apache Spark with Python and more

A TALE OF TWO LINKS

Both are Available Only in Microsoft Fabric

- ❖ Semantic Link

 - ❖ Base

 - ❖ Driver or API

 - ❖ Included in the default runtime for the current version

A LINK BETWEEN WORLDS

- ❖ Semantic Link Labs
 - ❖ Michael Kovalsky
 - ❖ “Kovalsky’s Laboratory”
 - ❖ “Expansion Pack” -- Kurt Buhler
 - ❖ Uses Semantic Link
 - ❖ `import sempy.fabric as fabric`
 - ❖ Open source – GitHub Repository
 - ❖ Active Development

SEMANTIC LINK LABS

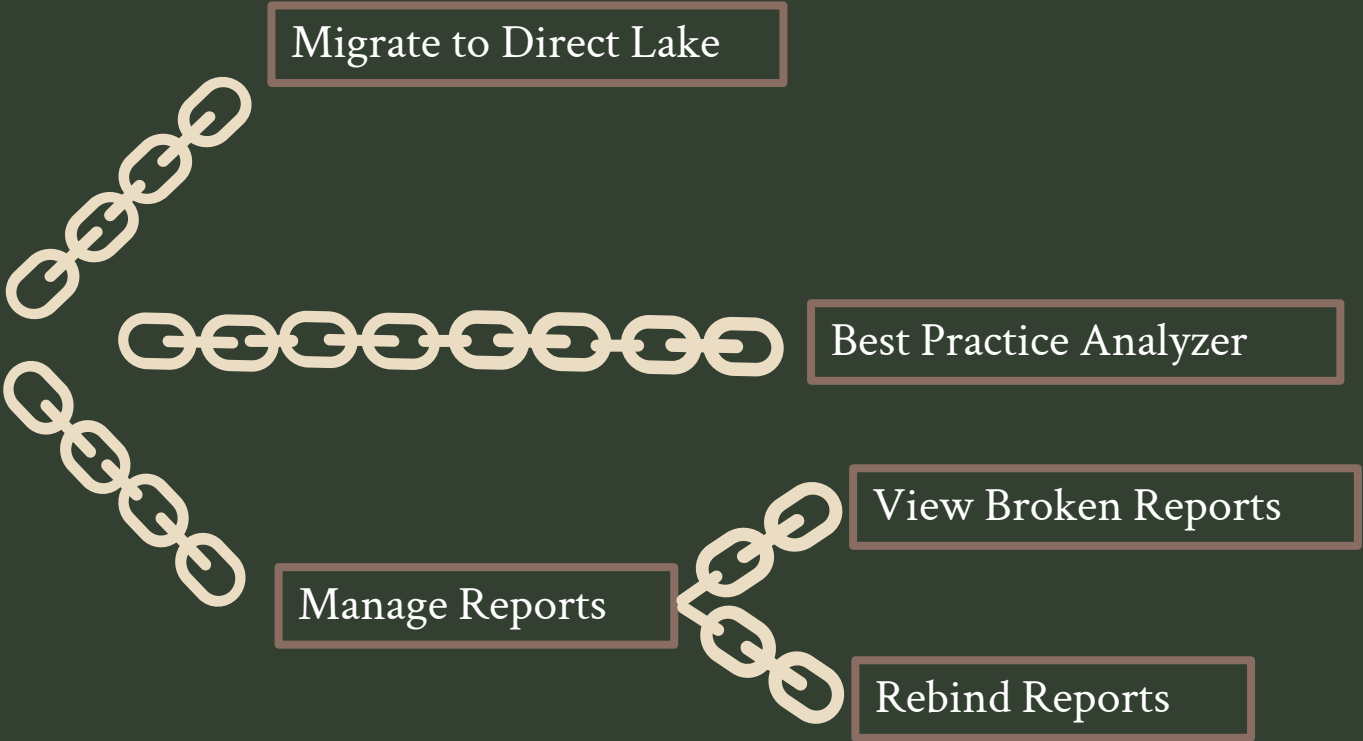
Microsoft Fabric



Semantic Link

- List Tables
- List Workspaces
- List Models
- List Reports

Semantic Link Labs



INSTALLING

- ❖ Semantic Link

- ❖ Spark 3.4 in default runtime

- ❖ Update to newest version

- ❖ %pip install -U semantic-link

- ❖ Spark 3.3 or below – need to install

- ❖ %pip install -U semantic-link

WHAT VERSION OF SPARK

Workspace settings

- General
- License info
- Azure connections
- System storage
- Git integration
- OneLake
- Workspace identity
- Network security
- Power BI
- Delegated Settings
- OneLake settings
- Data Engineering/Science**
- Spark settings**
- Data Factory

Spark settings

Configure and manage settings for Spark workloads and the default environment for the workspace.

Pool **Environment** High concurrency Automatic log

Set default environment

The default environment will provide Spark properties, libraries, and developer settings for notebooks and Spark job definitions in this workspace when users don't select a different environment. [Learn more about Set default environment](#)

Runtime

Runtime Version

Runtime version defines which version of Spark your Spark pool will use. [Learn more about Runtime Version](#)

1.2 (Spark 3.4, Delta 2.4)

SPARK VERSION COMMAND

spark.version

1 spark.version

✓ <1 sec - Command executed in 260 ms by Jason Romans on 2:00:03 PM, 10/08/24

'3.4.3.5.3.20240904.5'

WHAT VERSION OF SEMANTIC LINK

📦 %pip show semantic-link

📦 or %pip list | grep semantic-link

```
1 %pip show semantic-link
```

[23] ✓ 4 sec - Command executed in 3 sec 588 ms by Jason Romans on 11:49:10 PM, 10/19/24 PySpark (Python) ▾

```
... Name: semantic-link
Version: 0.8.1
Summary: Semantic link for Microsoft Fabric
Home-page: https://learn.microsoft.com/en-us/fabric/data-science/semantic-link-overview
Author: Microsoft
Author-email: semanticdatascience@service.microsoft.com
License: proprietary and confidential
Location: /nfs4/pyenv-dd0ba783-2069-4fdc-8c80-bc03c74db705/lib/python3.11/site-packages
Requires: semantic-link-functions-geopandas, semantic-link-functions-holidays, semantic-link-functions-meteostat, semantic-link-functions-phonenumbers, semantic-link-functions-validators, semantic-link-sempy
Required-by:
Note: you may need to restart the kernel to use updated packages.
```

IMPORT THE MODULE

Give it a friendly name – think Alias

```
import sumpy.fabric as fabric
```

DONE IN A NOTEBOOK

Code - Python

Text Descriptions - Markdown

Install the latest .whl package

Check [here](#) to see the latest version.

```
1 %pip install semantic-link-labs
```

- Session ready in 8 sec 603 ms. Command executed in 26 sec 202 ms by Jason Romans on 10:29:04 AM, 10/23/24

Install Wheel from File

```
1 %pip install /lakehouse/default/Files/semantic_link_labs-0.8.3-py3-none-any.whl
```

- Command executed in 22 sec 745 ms by Jason Romans on 10:07:06 AM, 10/23/24

Show Semantic Link and Labs installed

```
1 %pip show semantic-link-sempy
2 print('\n')
3 %pip show semantic-link-labs
```

✓ - Command executed in 6 sec 382 ms by Jason Romans on 10:30:20 AM, 10/23/24

Name: semantic-link-sempy

Version: 0.8.1

Summary: Semantic link for Microsoft Fabric

Home-page: <https://learn.microsoft.com/en-us/fabric/data-science/semantic-link-overview>

Author: Microsoft

USES PANDAS ECOSYSTEM (DATAFRAME)

Knowledge of working with
Pandas DataFrame helpful

DataFrame – data table



LIST SEMANTIC MODELS

```
import sempy.fabric as fabric  
fabric.list_datasets()
```

Dataset = Semantic Model

	Dataset Name	Dataset ID	Created Timestamp	Last Update
0	Contoso10K	47c34560-ef4d-46c6-825e-20cb9f11ba9d	2023-05-04 14:36:12	NaT
1	FabSLL_Lakehouse	e6d18d3a-b407-4fb3-813a-418b76388b11	2021-02-12 23:00:58	NaT
2	DataflowsStagingLakehouse	7c76f16d-2364-4c33-89bb-6960ac29cb5d	2021-02-12 23:00:58	NaT
3	DataflowsStagingWarehouse	625cb1b9-de7e-425f-90b3-727ac87268db	2021-02-12 23:00:58	NaT
4	Contoso10K_DL	261101fb-fc2d-4511-be34-1def1b4530fe	2019-09-17 05:50:29	NaT
5	Contoso10K_DLL1	40b52877-7e62-43cd-9a38-7c09b62f9048	2019-09-17 05:50:29	NaT
6	Contoso10K_ABC	cc93c0f1-e112-4adc-a009-3fb3a7a12f3b	2019-09-17 05:50:29	NaT
7	Contoso10K_SLL	e45a6ce5-5ea5-4b87-ad88-136dc4cab2c7	2019-09-17 05:50:29	NaT
8	Contoso10K_SL1	421040eb-ba74-4304-bb92-708c1667eb61	2019-09-17 05:50:29	NaT
9	Contoso10K_SL2	b84b790d-04a2-406d-ba17-2191186ebafa	2019-09-17 05:50:29	NaT
10	Contoso10K_SL9	5ca86b73-bef1-43d6-af09-5c12e358d391	2019-09-17 05:50:29	NaT

LIST TABLES

```
tables = fabric.list_tables(workspace="ws", dataset = "model")  
display(tables)
```

	Name	Description	Hidden	Data Category	Type
0	Customer		False		Table
1	Sales		False		Table
2	Date		False		Table
3	Store		False		Table
4	Product		False		Table

LIST WORKSPACES

1 `fabric.list_workspaces()`

✓ - Command executed in 806 ms by Jason Romans on 5:25:32 PM, 10/08/24

PySpark (Python) ▾

	Id	Is Read Only	Is On Dedicated Capacity	Capacity Id	Default Dataset Storage Format	Type	Name
0	bfab8dff-bdfc-4943-9996-7dc97a4e4d38	False	False	NaN	NaN	Workspace	JAXSQL2023
1	41e69008-f6c3-42c0-8c0a-739f1a7a9a0a	False	True	3b9ac229-bcbb-4aa4-8543-72b6db25e330	Small	Workspace	ONSQL2023
2	10c0ad3f-7a0b-4c0e-8b7d-f4a6170c5dde	False	True	3b9ac229-bcbb-4aa4-8543-72b6db25e330	Small	Workspace	SOFLSQL2023
3	ee7a8d30-f109-4848-9db3-be711a0f24d4	False	False	NaN	NaN	AdminInsights	Admin monitoring
4	10c2bd0a-5d3d-4e77-883c-f16af027fcd5	False	True	3b9ac229-bcbb-4aa4-8543-72b6db25e330	Small	Workspace	COLSQLSAT2023
5	365a3880-83f7-4014-bad3-a006c34e2bb1	False	True	3b9ac229-bcbb-4aa4-8543-72b6db25e330	Small	Workspace	BRSQSQLSAT2023
6	e6e71dc2-2a8e-482a-8bc9-ff5c16ed0311	False	True	3b9ac229-bcbb-4aa4-8543-72b6db25e330	Small	Workspace	DENSQLSAT2023

LIST WORKSPACES ON DEDICATED CAPACITY

```
1 ws = fabric.list_workspaces()
2 dedicated = ws[ws["Is On Dedicated Capacity"] == True]
3 display(dedicated)
```

✓ - Command executed in 790 ms by Jason Romans on 5:37:07 PM, 10/08/24

PySpark (Python) ▾

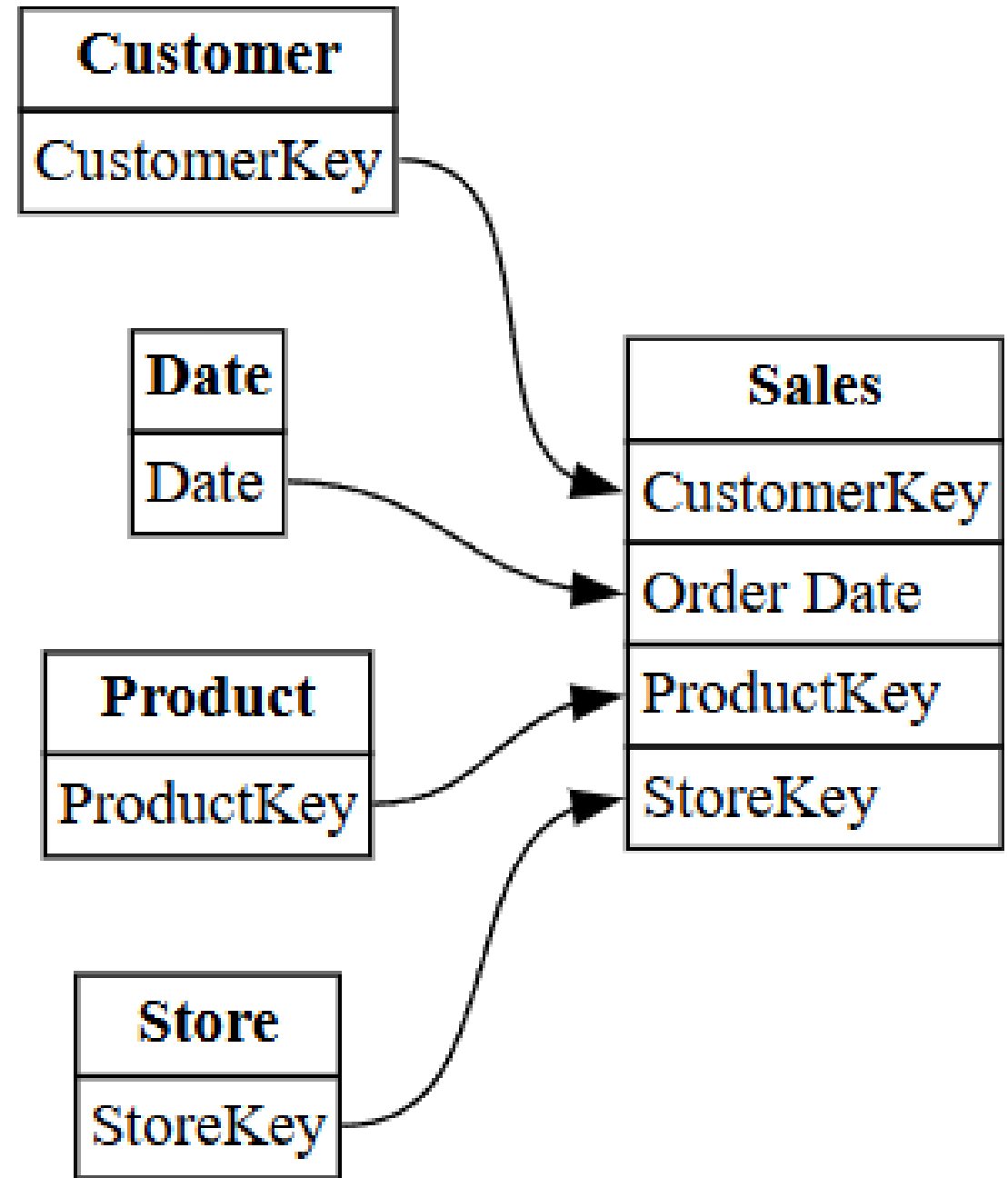
	Table	Chart	Download ▾		Showing rows 1 - 22		Inspect	Search	
	ABC Id	0/1 Is Read Only	0/1 Is On Dedicated Capacity	ABC Capacity Id	ABC Default Dataset Storage Format	ABC Type	ABC Name		
1	41e69008-f...	false	true	3b9ac229-bcbb...	Small	Workspace	ONSQL2023		
2	10c0ad3f-7...	false	true	3b9ac229-bcbb...	Small	Workspace	SOFLSQL2023		
3	10c2bd0a-...	false	true	3b9ac229-bcbb...	Small	Workspace	COLSQLSAT2023		
4	365a3880-...	false	true	3b9ac229-bcbb...	Small	Workspace	BRSQLSAT2023		
5	e6e71dc2-...	false	true	3b9ac229-bcbb...	Small	Workspace	DENSQSAT2023		
6	782b40a8-...	false	true	3b9ac229-bcbb...	Small	Workspace	SQLMAB		
7	62e33fa7-f...	false	true	3b9ac229-bcbb...	Small	Workspace	SQLSAT_Denver2023		
8	9a5d4f1a-b...	false	true	3b9ac229-bcbb...	Small	Workspace	MNSQSAT2023		
9	ba1e683d-...	false	true	3b9ac229-bcbb...	Small	Workspace	ORLANDOSQSAT2023		
10	be75ac57-...	false	true	3b9ac229-bcbb...	Small	Workspace	SV-SQLSat2023		
11	4d139fc8-0...	false	true	3b9ac229-bcbb...	Small	Workspace	ORWA23_SQLSAT		
12	23c86de6-...	false	true	3b9ac229-bcbb...	Large	Workspace	TE Training		
13	ad0afb41-...	false	true	3b9ac229-bcbb...	Small	Workspace	PremTest		
14	2e6ae09d-...	false	true	3b9ac229-bcbb...	Large	Workspace	SQLBits2024		
15	5f6564c6-1...	false	true	27046857-fc30-...	Small	Workspace	MetaDriven-2		

Selected Cell 14 of 21 cells

RELATIONSHIPS

```
from sempy.relationships import  
plot_relationship_metadata as prm
```

```
prm(  
    fabric.list_relationships(  
        workspace="Summit2025_SLL",  
        dataset="Hyrule10K"  
    )  
)
```



MODEL HEALTH IN MICROSOFT FABRIC

1. Best Practice Analyzer
2. Memory Analyzer
3. Community Notebooks

The screenshot displays the Microsoft Fabric user interface. At the top, a search bar and navigation tabs are visible. The 'Model health' dropdown menu is open, showing three options: 'Best practice analyzer', 'Memory analyzer', and 'Community notebooks'. Below the menu, a 'Share this data' section is visible, along with a table of data. The table has columns for 'Type', 'Relation', 'Location', 'Refreshed', 'Endorsement', and 'Sensitivity'. The first row of data shows 'Report', 'Downstream', 'AtlantaBI_SLL', '11/1/25, 9:07:33 ...', and empty cells for 'Endorsement' and 'Sensitivity'.

Type	Relation	Location	Refreshed	Endorsement	Sensitivity
Report	Downstream	AtlantaBI_SLL	11/1/25, 9:07:33 ...	—	—



SEMANTIC LINK DEMO



OUR JOURNEY



1. INTRO

2. NOTEBOOKS

3. SEMANTIC LINK

4. SEMANTIC LINK LABS

5. CONCLUSION

LEVEL 2: SEMANTIC LINK LABS

- ❖ Standard Python library
- ❖ Completely free and open-source
- ❖ Backed by Microsoft (GitHub repository)
- ❖ Continuous development and improvements

ORIGIN STORY

Migrate Semantic Models
from Import to Direct Lake

ACTIVE DEVELOPMENT

June 18, 2024 at 0.4.2

Now up to ~~0.12.5~~

~~0.12.6~~

0.12.7

INSTALL THE PACKAGE

Install Semantic Link Labs

```
%pip install semantic-link-labs
```

Import module with a shorter name

Easier to Type

```
import sempy_labs as labs
```

NOTEBOOKS TO GET YOU STARTED

- ❖ MIGRATION TO DIRECT LAKE
- ❖ BEST PRACTICE ANALYZER REPORT
- ❖ CAPACITY MIGRATION
- ❖ DELTA ANALYZER
- ❖ MODEL OPTIMIZATION
- ❖ QUERY SCALE OUT



EVEN MORE HELPER NOTEBOOKS

- ❖ REPORT ANALYSIS
- ❖ SQL
- ❖ SEMANTIC MODEL
MANAGEMENT
- ❖ SEMANTIC MODEL REFRESH
- ❖ SERVICE PRINCIPAL
- ❖ TABULAR OBJECT MODEL



OBSTACLES

May Require Spark

 This function may be executed in either a PySpark or pure Python notebook. If executing in a pure Python notebook, the DataFrame must be a pandas DataFrame.

May not fully work with Lakehouse Schemas

UPDATE DEPENDENCIES

Installing Semantic Link Labs
will update Semantic Link



EXPANDS SEMANTIC LINK

- ❖ Does not replace or overwrite
- ❖ Have both available
- ❖ You will use both
- ❖ Just like installing DuckDB or other packages

ENVIRONMENT

- ❖ Install shared libraries
- ❖ Spark settings
- ❖ Not available for
Pure Python Notebook

CREATE ENVIRONMENT

New item

☆ Favorites

☑ All items

Analyze and train data

Propose hypotheses, train models, and explore your data to make decisions and predictions.

Environment



Set up shared libraries, Spark compute settings, and resources for notebooks and Spark job definitions.



Develop data

Create and build your software, applications, and data solutions.

Environment



Set up shared libraries, Spark compute settings, and resources for notebooks and Spark job definitions.



NAME ENVIRONMENT

New Environment

×

Name

Create

Cancel

ADD LIBRARY

External repositories

Custom

Spark compute

Acceleration

Compute

Spark properties

Storage

Resources



There's nothing here yet

Add libraries from external repositories or via a YML

ADD LIBRARY FROM REPO (CHOICE)

Add library from public repository

 When you add a library from a new source, the repository address will be added to the channel address. You can edit the order of the repositories in the YML editor view. 

Public repository

☒ Pip

☐ Conda

Name ^{*}

semantic-link-labs

Version

0.12.7 

Add

Cancel

SPECIFIC VERSIONS

- ❖ Choose your development version
- ❖ Control your upgrade path

LIST OF LIBRARIES

1 You have unpublished changes. To apply these changes to notebooks and Spark job definition run in this environment, select Publish. To save your changes without updating the environment, select Save.

Libraries

 Built-in

 External repositories

1

 Custom

Spark compute

 Acceleration

 Compute

- Spark properties

Libraries from external repositories

Add libraries from public or private repositories using the list view or via YML. Added libraries will be available if you run your notebook or Spark job

☐	Library 	Version	Status	Last updated
☐	semantic-link-labs	0.12.7	New	New

PUBLISH (IMPORTANT)

 You have unpublished changes. To apply these changes to notebooks and Spark job definition run

Save Publish

Libraries

 Built-in

 External repositories **1**

 Custom

Spark compute

 Acceleration

 Libraries from external repositories

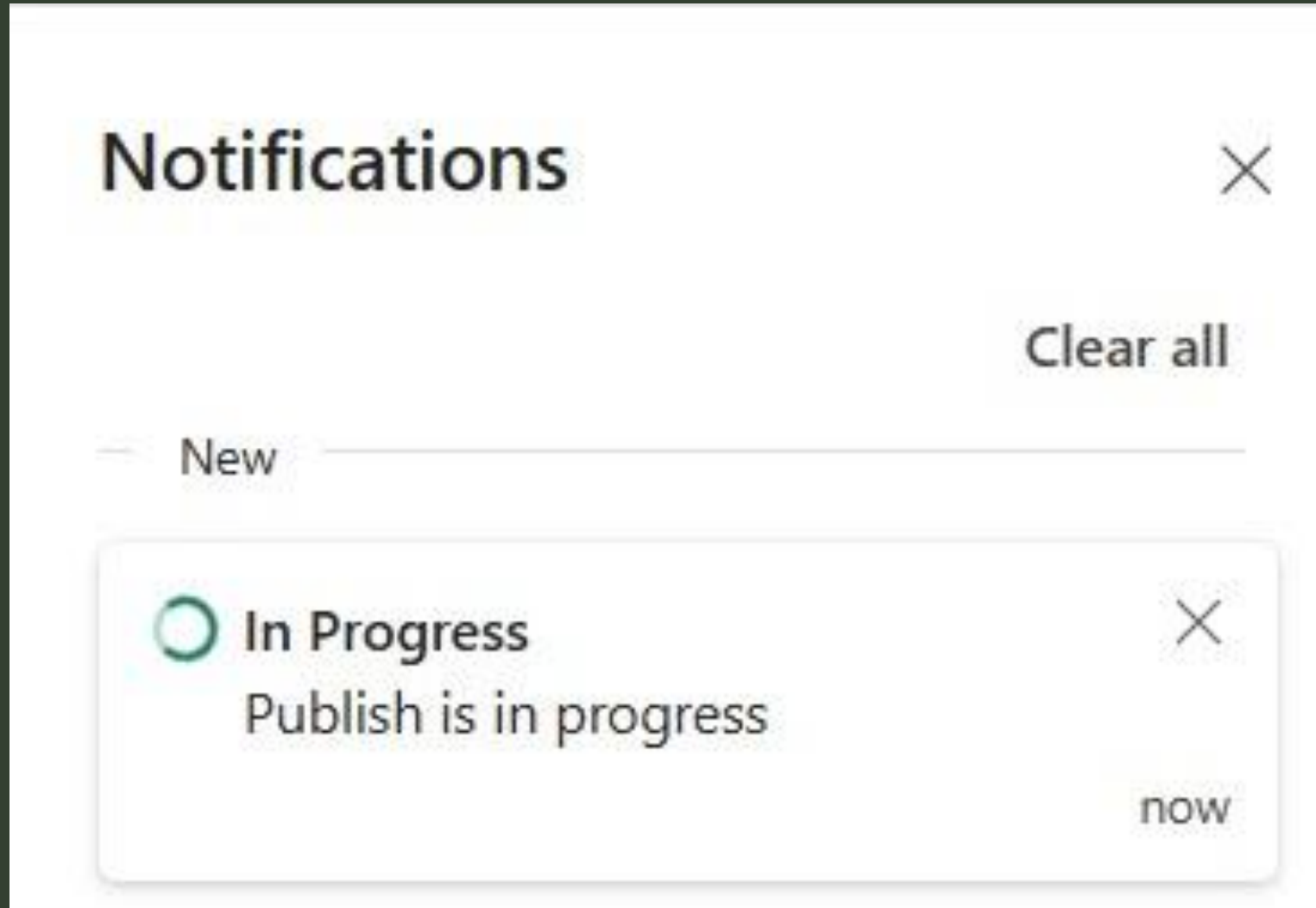
Add libraries from public or private repositories to your environment. [Learn more](#)

☐ Library ↑

☐ semantic-link-labs

 Filter by name

ENVIRONMENT TAKES TIME TO PUBLISH



WORKSPACE SETTINGS - ENVIRONMENT

Workspace settings

- General
- License info
- Azure connections
- System storage
- Git integration
- OneLake
- Workspace identity
- Network security

Power BI

Delegated Settings

Data
Engineering/Science

Spark settings

Data Factory

This section contains unsaved changes.

Spark settings

Configure and manage settings for Spark workloads and the default environment for the workspace.

Pool **Environment** Jobs High concurrency Automatic log

Set default environment

On

The default environment will provide Spark properties, libraries, and developer settings for notebooks and Spark job definitions in this workspace when users don't select a different environment. [Learn more about Set default environment](#)

Semantic_Link_Labs

Filter by keyword

Available environments

✓ Semantic_Link_Labs

Runtime: 1.3 (Spark 3.5, Delta 3.2), Compute: Medium, 1-10 nodes



New environment

Spark driver memory
56g

Spark executor core
8

Spark executor memory
56g

Dynamically allocate executors
Enabled

Spark executor instances
1-9

SET ENVIRONMENT - NOTEBOOK SETTING

The screenshot displays the Microsoft Fabric notebook environment. At the top, the 'Fabric' logo is on the left, followed by a tab labeled '2.HelperNotebooks' and a 'Saved' status indicator. A search bar is located on the right. Below the top bar, a navigation menu includes 'Home', 'Edit', 'AI tools', 'Run', and 'View'. A secondary toolbar contains icons for saving, downloading, settings, running all code, connecting, and selecting the environment (currently 'PySpark (Python)').

The left sidebar features an 'Explorer' panel with 'Data items' and 'Resources' sections. The 'Data items' section has an 'Add data items' button. The main workspace shows a code cell with the command `%pip install -U semantic-link`. Below the code, a status message reads: '- Session ready in 13 sec 257 ms. Command executed'.

Two context menus are open. The first, titled 'Environment', is positioned over the 'Environment' button and lists: 'Workspace default', 'Workspace Settings' (with details: Runtime: 1.3 (Spark 3.5, Delta 3.2), Compute: Medium, 1-10 nodes), 'Change environment', and 'New environment'. The second menu, titled 'Workspace default', is open over the 'Workspace default' option and lists: 'Workspace: Summit2025_SLL', 'Open environment', 'Compute', 'Dynamically allocate true', and 'Number of executors 1 - 9'.

EXECUTING NOTEBOOK USING A PIPELINE

Either

Use Environment with Semantic Link Labs

-  Microsoft recommended way

-  Only for Spark

Python inline installation

-  Enable %pip install for pipeline, add "_inlineInstallationEnabled" as bool parameter equals True in the notebook activity parameters.

Reference:

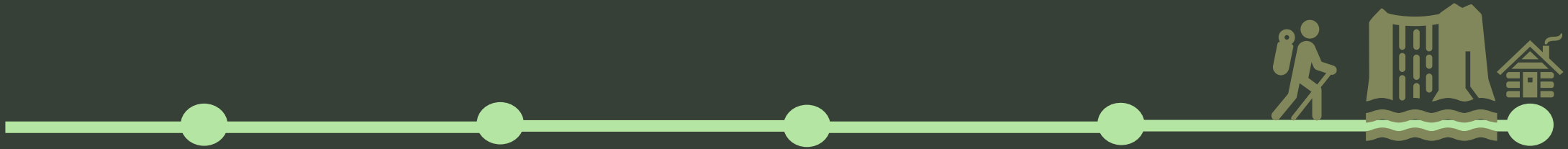
-  <https://learn.microsoft.com/en-us/fabric/data-engineering/library-management>



SEMANTIC LINK LABS DEMO



OUR JOURNEY



1. INTRO

2. NOTEBOOKS

3. SEMANTIC LINK

4. SEMANTIC LINK LABS

5. CONCLUSION

NOT IF BUT WHEN

- ❖ Your go-to solution (sooner or later)
- ❖ Ever-expanding world of capabilities
- ❖ Evolving to adapt to Fabric changes
 - ❖ For Example: Lakehouse Schemas

RESOURCES

Semantic Link

- 📖 <https://learn.microsoft.com/en-us/fabric/data-science/semantic-link-overview>

Semantic Link labs

- 📖 <https://github.com/microsoft/semantic-link-labs>

📖 Semantic Link Labs – Read the Docs

- 📖 https://semantic-link-labs.readthedocs.io/en/stable/sempy_labs.html

RESOURCES

Tabular Editor Training (Notebooks, Semantic Link, and semantic-link-labs)

❖ <https://tabulareditor.com/learn>

Sandeep Pawar

❖ <https://fabric.guru/fabric-semantic-link-and-use-cases>

Zelda Dataset (Zelda: Breath of the Wild Hyrule Compendium)

❖ <https://www.kaggle.com/datasets/elisamork/zelda-breath-of-the-wild-hyrule-compendium>

THANK YOU

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